

Grade 10 Biology: Principles of Inheritance and Genetics

1. Introduction to Genetics

Genetics is the branch of biology concerned with the study of heredity, genes, and variation in living organisms. Gregor Mendel

Key Terminology

- Gene: Functional unit of heredity that determines a specific trait.
- Alleles: Alternative forms of a gene (e.g., T for Tall, t for Dwarf).
- Phenotype: Observable physical manifestation of a trait (e.g., Purple flowers).
- Genotype: Genetic makeup of an organism (e.g., TT, Tt, or tt).
- Homozygous: Having identical alleles for a single trait (TT or tt).
- Heterozygous: Having two different alleles for a trait (Tt).

2. Mendel's Laws of Inheritance

1. Law of Dominance: In a heterozygous pair of alleles, the dominant allele suppresses the expression of the recessive allele.
2. Law of Segregation: During gamete formation, the two alleles of a gene separate so that each gamete carries only one allele.
3. Law of Independent Assortment: Alleles of two or more different genes assort independently of one another during gamete formation.

3. Monohybrid Cross Example

Crossing a pure tall pea plant (TT) with a pure dwarf pea plant (tt):

- Parents (P1): $TT \times tt$
- F1 Generation: All offspring are Tt (Phenotype: 100% Tall)
- F2 Generation (Selfing $Tt \times Tt$):
 - Genotypic Ratio: $1 \text{ } TT : 2 \text{ } Tt : 1 \text{ } tt$ (1:2:1)
 - Phenotypic Ratio: $3 \text{ } \text{Tall} : 1 \text{ } \text{Dwarf}$ (3:1)

4. Assessment Questions

1. Differentiate between genotype and phenotype.
2. State Mendel's Law of Segregation with an example.
3. Using a Punnett square, determine the probability of obtaining dwarf pea plants when two heterozygous tall pea plants (Tt) are crossed.